

CPCS 202 – Programming I

03 – Selection Statements

Objectives

- Combinations of *if*, *else* and *else if* statements
- Nested *if*
- *switch-case* statements
- Tertiary *if* (Conditional Expression)

If statement - Structure

```
if (boolean expression) {  
    .....  
}
```

if statement – Pseudo code

```
1: print "Enter a number"
2: read number
3: IF number divisible by 2 AND 3
4:     print "Number divisible by 6"
5: ENDIF
6: print "Thank you!"
```

If statement - Java

```
System.out.print("Enter a number : ");  
int number = in.nextInt();  
if ((number%2==0) && (number%3==0)) {  
    System.out.println("Number divisible by 6");  
}  
System.out.println("Thank you!");
```

Writing styles - if statements

Single statements - Can be written inside curly braces, as

```
if (boolean expression) {  
    .....  
}
```

or without curly braces, as

```
if (boolean expression)  
    .....
```

Multiple statements have to be written inside curly braces

```
if (boolean expression) {  
    .....  
    .....  
    .....  
}
```

If-else statement - Structure

```
if (boolean expression ) {  
    .....  
}  
else {  
    .....  
}
```

If-else statement – Pseudo code

```
1: print "Enter a number"
2: read number
3: IF number divisible by 2 AND 3
4:     print "Number divisible by 6"
5: ELSE
6:     print "Number not divisible by 6"
7: ENDIF
8: print "Thank you!"
```

If-else statement - Java

```
System.out.print("Enter a number : ");
int number = in.nextInt();
if ((number%2==0) && (number%3==0)) {
    System.out.println("Number divisible by 6");
}
else {
    System.out.println("Number not divisible by 6");
}
System.out.println("Thank you!");
```

If-else if-else statement - Structure

```
if (boolean expression #1) {  
    .....  
}  
else if (boolean expression #2) {  
    .....  
}  
else {  
    .....  
}
```

If-else if-else statement – Pseudo code

```
1: print "Enter a number"
2: read number
3: IF number divisible by 2
4:     print "Number divisible by 2"
5: ELSE IF number divisible by 3
6:     print "Number divisible by 3"
7: ELSE
8:     print "Number not divisible by 2 or 3"
9: ENDIF
10: print "Thank you!"
```

If-else if-else statement - Java

```
System.out.print("Enter a number : ");
int number = in.nextInt();
if (number%2==0) {
    System.out.println("Number divisible by 2");
}
else if (number%3==0) {
    System.out.println("Number divisible by 3");
}
else {
    System.out.println("Number not divisible by 2 or 3");
}
System.out.println("Thank you!");
```

Nested if - Structure

```
if (boolean expression #1) {  
    if (boolean expression #2) {  
        .....  
    }  
    else {  
        .....  
    }  
}
```

Nested If – Pseudo code

```
1: print "Enter a number"
2: read number
3: IF number divisible by 2
3:     IF number divisible by 3
4:         print "Number divisible by 6"
5:     ELSE
6:         print "Number not divisible by 6"
7:     ENDIF
8: ENDIF
9: print "Thank you!"
```

Nested if statement - Java

```
System.out.print("Enter a number : ");
int number = in.nextInt();
if (number%2==0) {
    if (number%3==0) {
        System.out.println("Number divisible by 6");
    }
    else {
        System.out.println("Number not divisible by 6");
    }
}
System.out.println("Thank you!");
```

Tertiary If statement (Conditional Expression)

variable = (boolean expression) ? Stmt #1 : Stmt #2;

Example

```
y = (x > 10) ? 3 : 5;
```

```
if (x>10) {  
    y = 3;  
}  
else {  
    y = 5;  
}
```

Switch-case Statement

Consider the following if statement

```
if (day==1)
    System.out.println("Sunday");
else if (day==2)
    System.out.println("Monday");
else if (day==3)
    System.out.println("Tuesday");
else
    System.out.println("Other day");
```

... can alternatively be written as

```
switch(day) {  
    case 1:  
        System.out.println("Sunday");  
        break;  
    case 2:  
        System.out.println("Monday");  
        break;  
    case 3:  
        System.out.println("Tuesday");  
        break;  
    default:  
        System.out.println("Other days");  
}
```

Example #1 – problem statement

Read three numbers from the user and print them in ascending order

Sample input output

Input	Output
24, 15, 83	15, 24, 83
56, 98, -54	-54, 56, 98

Example #1 – Java code

```
int A = in.nextInt();
int B = in.nextInt();
int C = in.nextInt();
if (A<B && B<C)
    System.out.println(A + ", " + B + ", " + C);
else if (A<C && C<B)
    System.out.println(A + ", " + C + ", " + B);
else if (B<A && A<C)
    System.out.println(B + ", " + A + ", " + C);
else if (B<C && C<A)
    System.out.println(B + ", " + C + ", " + A);
else if (C<A && A<B)
    System.out.println(C + ", " + A + ", " + B);
else if (C<B && B<A)
    System.out.println(C + ", " + B + ", " + A);
```

Example #2 – problem statement

Read a Hex digit from the user and print its corresponding decimal digit

Sample input output

Input	Output
5	5
A	10
C	11
K	Not a Hex Digit

Example #2 – Java Code

```
char digit = in.next().charAt(0);
if (digit >= '0' && digit <= '9') {
    System.out.println(digit);
}
else if (digit >= 'A' && digit <= 'F') {
    System.out.println(digit-'A'+10);
}
else {
    System.out.println("Not a Hex Digit");
}
```

Example #3 – problem statement

Read a character from the user and print whether the character is lower case or upper case

Sample input output

Input	Output
b	Lower case
G	Upper case
%	Not a letter

Example #3 – Java Code

```
char letter = in.next().charAt(0);

if (letter >= 'a' && letter <= 'z') {
    System.out.println("Lowercase (small) letter");
}
else if (letter >= 'A' && letter <= 'Z') {
    System.out.println("Uppercase (capital) letter");
}
else {
    System.out.println("Not a letter");
}
```

Example #4 – problem statement

Write a program that simulates a simple Math Learning Tool. It should provide a menu for Addition, Subtraction and Multiplication. For each operation chosen by the user, it should randomly generate two integers and should then ask the user to input the result of the operation on the two numbers. It should then display a message stating if the answer is correct or wrong and then display the correct answer.

```
System.out.println("1. Addition");
System.out.println("2. Subtraction");
System.out.println("3. Multiplication");
System.out.print("Please enter your choice : ");
char choice = in.next().charAt(0);

int a = (int)(100 + Math.random()*900);
int b = (int)(100 + Math.random()*900);

switch (choice) {
    case '1':
        System.out.print("What is " + a + " + " + b + " = ");
        int ans = in.nextInt();
        int correctAns = a + b;
        if (ans == correctAns) {
            System.out.println("Congratulations!!! You win a Toyota Camry!!!");
        }
        else {
            System.out.println("Sorry. Please try again!");
            System.out.println("The correct answer is " + correctAns);
        }
        break;
```

```
case '2':  
    // Swap the numbers if 2nd number larger than 1st to avoid a negative answer  
    if (b>a) {  
        int temp = a;  
        a = b;  
        b = temp;  
    }  
    System.out.print("What is " + a + " - " + b + " = ");  
    int ans = in.nextInt();  
    int correctAns = a - b;  
    if (ans == correctAns) {  
        System.out.println("Congratulations!!! You win a BMW!!!");  
    }  
    else {  
        System.out.println("Sorry. Please try again");  
        System.out.println("The correct answer is " + correctAns);  
    }  
    break;
```

```

case '3':
    // Change level of difficulty by using a single digit number
    a = (int)(1 + Math.random()*9);
    b = (int)(1 + Math.random()*9);

    System.out.print("What is " + a + " * " + b + " = ");
    int ans = in.nextInt();
    int correctAns = a * b;
    if (ans == correctAns) {
        System.out.println("Congratulations!!! You win a Ferrari!!!");
    }
    else {
        System.out.println("Sorry. Please try again");
        System.out.println("The correct answer is " + correctAns);
    }
    break;
default:
    System.out.println("Sorry!! Wrong choice entered!!");
}

```

Usage : System.out.printf()

```
double a = 236.0/13, b = 234.0/17;  
int x = 89;  
System.out.printf("x is %d, a is %.2f\n", x, a);  
System.out.printf("b is %.4f\n", b);
```

Output

```
x is 89, a is 18.15  
b is 13.7647
```

Usage : System.out.printf()

```
double a = 236.0/13, b = 22234.0/17;  
System.out.printf("Eggs   %8.2f\n", a);  
System.out.printf("Juice  %8.2f\n", b);
```

Output

```
Eggs    18.15  
Juice  1307.88
```

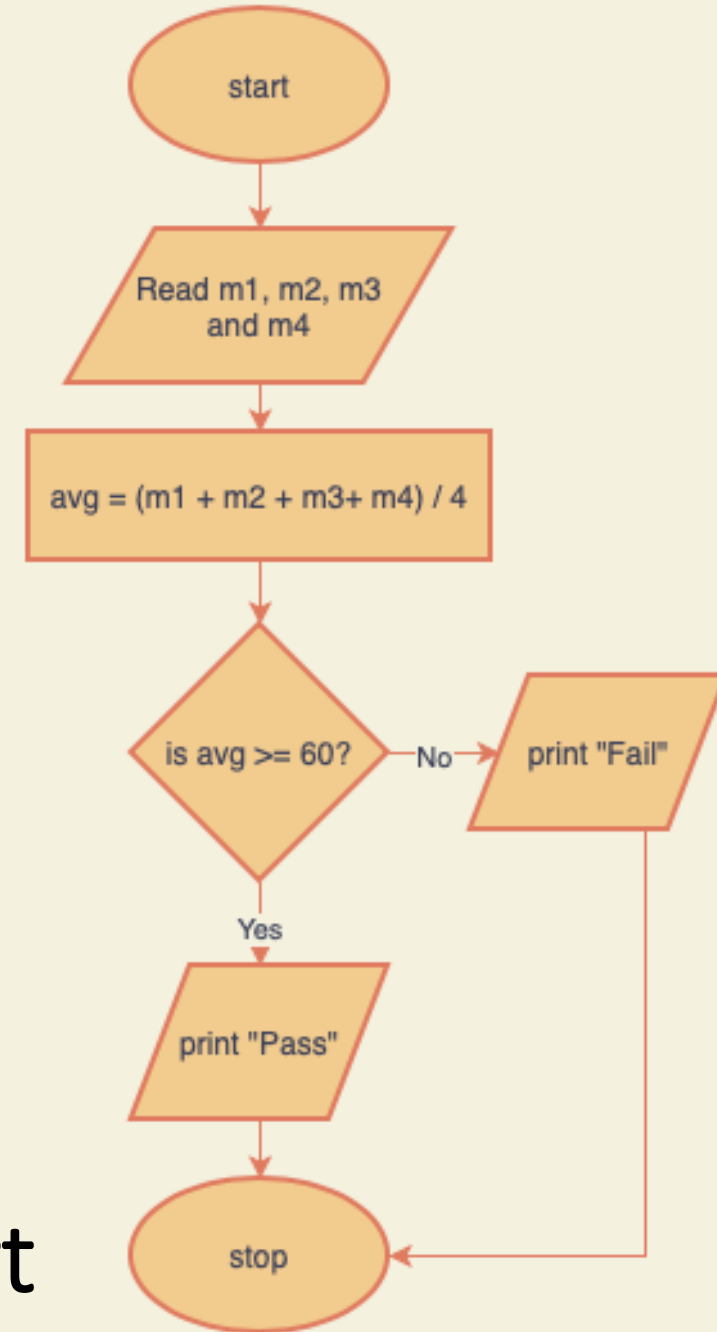
Examples for Practice

Translate the given problem statement, pseudocode or flowchart into its equivalent Java code

Example #1- Pseudocode

```
1: read m1, m2, m3 and m4
2: average = (m1+m2+m3+m4) / 4
3: IF average >=60
4:     print "Pass"
5: ELSE
6:     print "Fail"
7: ENDIF
```

Flowchart



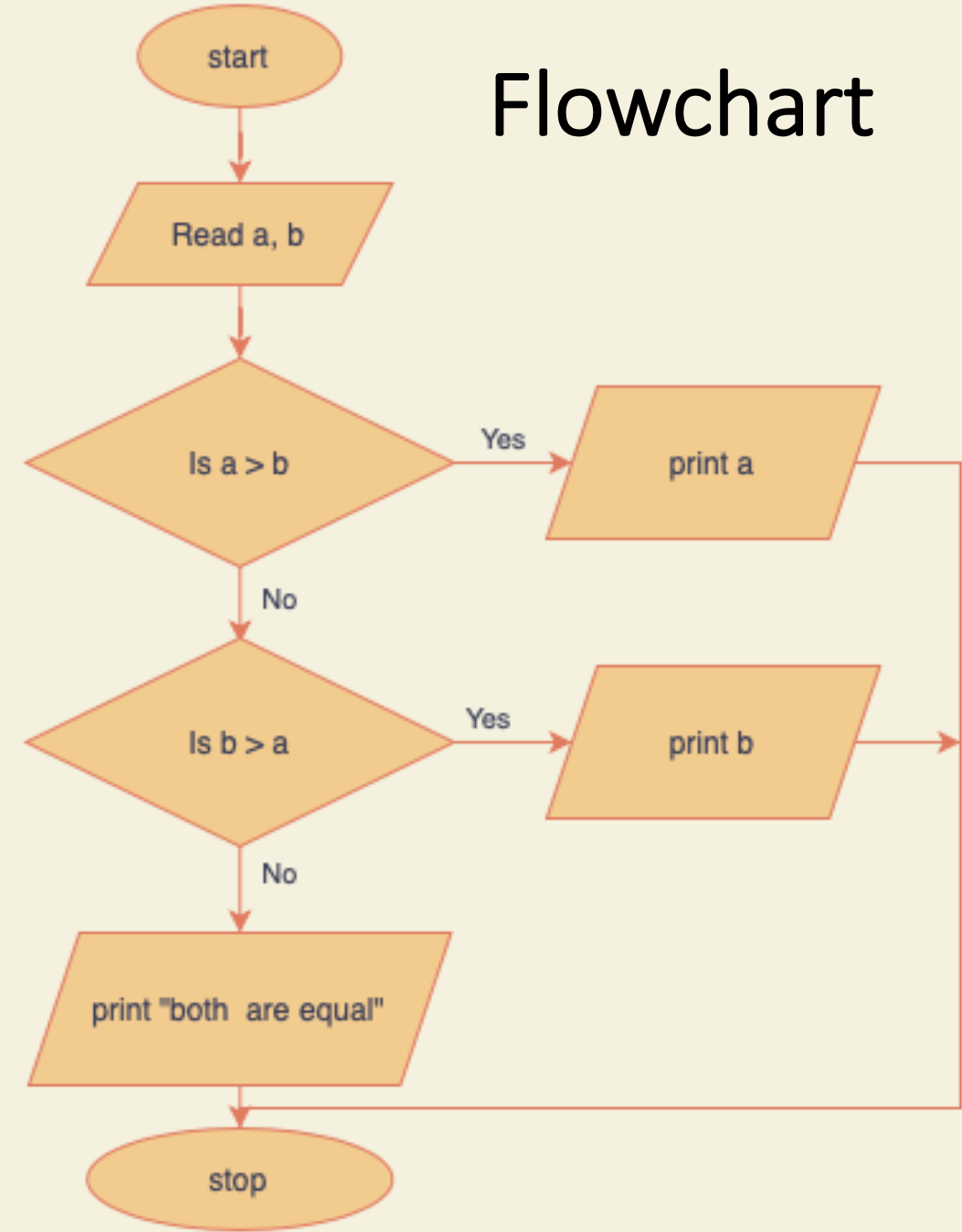
Example #2 – Larger of two numbers

Read two numbers from the user and then display the larger of the two numbers. If both are equal then display a message that both numbers are equal.

Example #2 - Pseudocode

```
1: read a, b
2: IF a > b
3:   print a
4: ELSE IF b > a
5:   print b
6: ELSE
7:   print "both are equal"
8: ENDIF
```

Flowchart



Example #3 – Pooja and ATM

Pooja would like to withdraw X \$US from an ATM. The cash machine will only accept the transaction if X is a multiple of 5, and Pooja's account balance has enough cash to perform the withdrawal transaction (including bank charges). For each successful withdrawal, the bank charges 0.50 \$US. Calculate Pooja's account balance after an attempted transaction. (Source : <https://www.codechef.com/problems/HS08TEST>)

Sample input and output

Input	Output	Explanation
30 120.00	89.50	Successful Transaction
42 120.00	120.00	Unsuccessful transaction. Amount not multiple of 5
300 120.00	120.00	Unsuccessful transaction. Insufficient balance